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SOC and SOH Prediction of Lithium-Ion Batteries Based on LSTM–AUKF Joint Algorithm

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ABSTRACT

Lithium batteries are increasingly favored for energy storage due to their high energy density, long cycle life, and robust charge and discharge rates. However, safety concerns necessitate the implementation of a battery management system (BMS) to monitor battery status, maintain energy balance, and provide failure warnings to ensure safe operation. This paper proposes an efficient BMS for high-voltage, high-current lithium battery energy storage. The approach leverages a multihead-attention-enhanced long short-term memory (LSTM) neural network combined with an adaptive unscented Kalman filter to accurately calculate the battery's state of charge (SOC) and state of health (SOH). To improve accuracy, various factors such as temperature and internal resistance were considered. The algorithm was validated through hardware and simulation experiments, with experimental data compared to estimation results to demonstrate its precision. The findings show strong convergence and tracking capabilities, with SOC estimation presenting a maximum error of 1.5% and SOH estimation a maximum error of under 0.4%. We expect that this approach will allow for a more refined evaluation of SOC and SOH in lithium-ion batteries, potentially improving Li-ion battery system management.

1 | Introduction

According to the World Energy Statistics Yearbook, 2022, fossil fuels accounted for 82% of global primary energy consumption in 2021, with oil, natural gas, and coal accounting for 31.6%, 23.5%, and 26.7%, respectively [1]. This overdependence on fossil fuels, a nonrenewable resource, not only poses the threat of an energy crisis due to dwindling reserves but also contributes to increased carbon dioxide emissions, resulting in increased greenhouse effect and ecological imbalance. Consequently, there has been a shift toward the usage of clean energy sources. Energy storage systems play a key role in clean energy adoption. Electrochemical energy storage systems offer advantages such as precise control, easy installation and expansion, and quick response to changes [2, 3]. Among various energy storage systems proposed and developed, lithium-ion (Li-ion) batteries are frequently chosen by electric vehicle manufacturers due to their enticing attributes, including high energy density, high power density, extended lifespan, and low self-discharge rates. Although Li-ion batteries surpass the performance of traditional lead-acid batteries, they are susceptible to environmental and operational conditions and must be maintained within an optimal range. To prevent Li-ion batteries from operating under extreme conditions, an advanced battery management system (BMS) is essential for monitoring the state of the battery [4].

Within the context of BMS, estimating the internal state presents a formidable challenge. Over the years, extensive research has been conducted on two pivotal states: the state of charge (SOC) and the state of health (SOH) of batteries. Estimation of the state of charge and state of health has consistently been a significant challenge for devices powered by batteries [5]. Due to the inherent characteristics of batteries' nature, battery capacity diminishes with each charging and discharging cycle. The decline in battery health capacity exhibits a nonlinear behavior that traditional modeling methods cannot accurately represent.

The core state of batteries is intertwined, highlighting the limitation of estimating the SOC and SOH separately; thus, the necessity for a coupled joint estimation arises. Current methodologies for joint SOC and SOH estimation fall into two primary categories: model-based approaches and data-driven strategies [6]. The model-based estimation predominantly utilizes equivalent circuit models (ECM) and employs various filtering algorithms to estimate battery states and aging parameters, including extended Kalman filter (EKF) [7], dual extended Kalman filter (DEKF) [8], unscented Kalman filter (UKF) [9], and dual unscented Kalman filter (DUKF) [10], among others. The accuracy of model-based SOC/SOH estimation heavily relies on the precision of the model. Inaccuracies in the battery model or its parameters compromise estimation accuracy. While complex models can describe battery internal dynamics more accurately, enhancing estimation precision, they also lead to increased computational demand. Balancing the relationship between model complexity, estimation accuracy, and computational speed remains an unresolved challenge in model-based joint estimation. Conversely, data-driven joint estimation leverages machine learning to capture the changing characteristics of SOC and SOH, independent of precise models or decoupling [11–14]. This approach generally achieves higher estimation accuracy compared to model-based methods but requires time-consuming data collection, and the quality of training data directly impacts the accuracy of estimations.

Model-based joint estimation heavily relies on the accuracy of the model itself, whereas data-driven estimation depends on the availability and quality of training data, often incurring significant computational costs. This poses a challenge for BMS, which are typically deployed on edge computing devices, where computational resources are limited. This contradiction emphasizes the need for optimizing the balance between model precision and computational efficiency, particularly in environments constrained by hardware capabilities. Such optimization is crucial for ensuring that BMS can operate effectively, making real-time decisions without compromising on reliability or performance due to the inherent limitations of edge computing platforms.

In addressing the challenges identified in current research, further consideration is given to the dynamic relationship between SOC and SOH in batteries. It is recognized that SOH predominantly determines the capacity within SOC equations on a broader temporal scale, while at finer time scales, the SOC has a notable impact on the rate of SOH change. Acknowledging the mutual influence between SOC and SOH, an approach to decouple their interrelated dynamics has been adopted. This involves the deployment of two separate estimation algorithms for updating SOC and SOH independently, with each estimator reciprocally refreshing the state and parameters of the other. The method for determining remaining battery capacity directly leverages the relationship between current and voltage, a model that is both straightforward to implement and robust [15], thus advocating for a model-based approach for SOC estimation. Conversely, developing a robust battery aging model is fraught with challenges due to the intricate interplay of factors such as temperature and depth of discharge [16].

Consequently, data-driven approaches, which facilitate the decoupling of these factors, have garnered significant attention [17]. Zhang et al. [18], Zhu et al. [19], and Li et al. [16] have proposed the use of machine learning algorithms to accurately predict the SOH by analyzing the electrochemical impedance spectroscopy (EIS) of batteries. However, the acquisition of EIS data requires specialized equipment, which is expensive and not practical for widespread application. To address these limitations, recent research has focused on extracting relevant features from partial charge and discharge curves and employing these features for SOH prediction. For instance, Che et al. [20, 21] have demonstrated reliable SOH predictions across varying temperatures and operating conditions using this approach. However, in the context of embedded devices, the use of neural networks for feature extraction imposes a significant burden on computational resources. As a result, we proposed a hybrid approach that leverages the strengths of both methodologies. Specifically, we employ a model-based method to extract feature parameters and propose a long short-term memory (LSTM) [22, 23] associated with an adaptive unscented Kalman filter (AUKF) [24] fusion algorithm for SOC and SOH estimation. Furthermore, we have developed a corresponding hardware platform to validate our approach. The primary contributions of this paper are as follows.

1. This paper employs the AUKF as the estimation algorithm for the SOC of lithium-ion batteries, offering an advantage over the Extended Kalman Filter by presenting less dependence on the system's degree of nonlinearity. Additionally, the research integrates a noise covariance selection with a dual-polarization equivalent circuit model within this algorithm, establishing system state space equations based on the dynamic characteristics of lithium batteries. Essential charging and discharging experiments on lithium batteries were conducted, and the algorithm was utilized to identify both the static characteristics and dynamic polarization parameters of the lithium batteries.
2. Addressing the complex challenges in estimating the SOH of batteries—characterized by multiple influencing factors and significant nonlinearity—this article presents an enhanced LSTM neural network integrated with a multihead attention mechanism. This advanced architecture is designed to estimate the current maximum available capacity of batteries by accounting for factors such as battery aging and internal resistance, while also enabling real-time parameter updates within the SOC estimation algorithm. By incorporating an adaptive update process for both the gain and observation covariance, the approach not only improves estimation accuracy but also reduces computational demand and mitigates the impact of unknown noise. The multihead attention mechanism further enhances the model's ability to focus on the most relevant aspects of the input data, leading to more precise

parameter estimation. Additionally, the adaptability of the algorithm allows for real-time updates of parameters in the state equation based on SOH estimation, considering factors like battery aging and capacity decay during the SOC estimation process, thereby increasing the reliability of the results.

3 In this paper, a dedicated testing platform for Battery Management system has been developed to validate the accuracy of the proposed framework. Through rigorous experimentation, as detailed in our study, it has been demonstrated that this algorithm plays a pivotal role in mitigating the impact of external variables and nonlinear behaviors that often plague battery SOH estimations. The significance of these findings cannot be overstated, as they directly contribute to a marked enhancement in both the reliability and precision of SOH assessments for batteries.

2 | Methods

2.1 | Overall Pipeline

By analyzing existing SOC and SOH estimation algorithms, in this paper, we proposed the LSTM neural network method for SOH estimation, considering parameters such as temperature, battery aging, and internal resistance to estimate the current maximum capacity of the battery. We employed the AUKF used for SOC estimation to dynamically update noise and observation covariances, thereby ensuring higher SOC estimation accuracy with fewer computational operations and reducing the influence of noise on the estimation results [25]. This joint estimation method considers factors such as battery aging and capacity degradation in the process of SOC estimation so that the estimation results are closer to the real situation. The overall flow of the algorithm is shown in Figure 1.

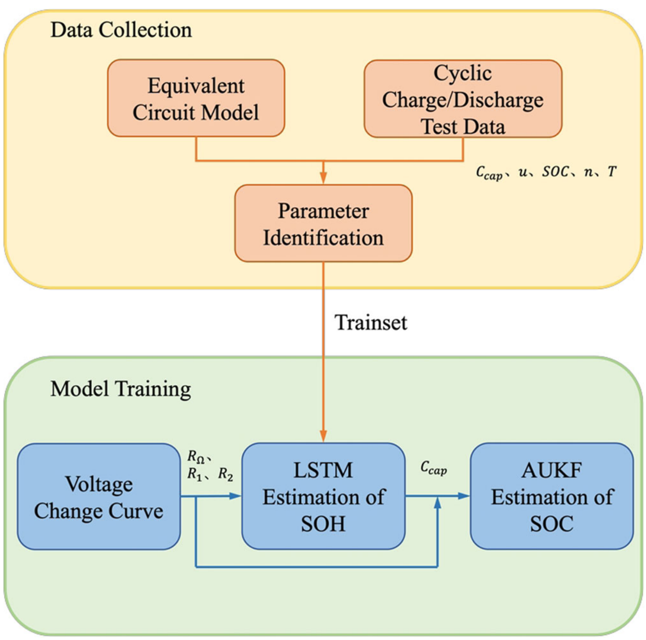

FIGURE 1 | Overview of the proposed LSTM–AUKF battery management system.

We established a suitable equivalent circuit model for the Li-ion battery, derived the state space equations, determined the parameters through discharge tests, and normalized the parameters. In the proposed LSTM neural network, ohmic resistance, polarization resistance, number of charging and discharging cycles, and temperature are used as inputs, and the maximum available capacity is the output and is used to train the model.

Before the end of each charging operation, when the system monitors that the charger is disconnected or the BCU charging relay is disconnected, it starts recording the battery terminal voltage conditions during the 10-min quiescent-state period after the end of charging, fits the voltage curve, calculates the ohmic and polarization resistances, measures the ambient temperature, and reads the number of battery charging and discharging cycles from the storage module. These data are then inputted into the proposed neural network model to predict the current maximum capacity of the battery, based on which the battery SOH is determined.

The maximum available capacity C_{cap}, ohmic resistance R_{Ω}, polarization resistances R_1 and R_2, and polarization capacitances C_1 and C_2 obtained using the LSTM model are substituted into the model equations, the model is updated, the AUKF is applied to estimate the battery SOC, and the obtained SOC value is recorded in the storage chip.

2.2 | Equivalent Circuit Model

Considering the complex internal characteristics of lithium batteries, we designed an equivalent circuit model to describe their external behavior [26, 27]. This model is used to establish the battery's state-space equation in the AUKF SOC estimation algorithm. Striking a balance between complexity and accuracy, we selected the dual polarization equivalent circuit, as illustrated in Figure 2.

By using the set of equations for the two ends of the circuit according to Kirchhoff's law and discretizing it, the following relationship can be obtained:

$$\begin{cases} U_1(k+1) = e^{\frac{-\Delta t}{R_1 c_1}} U_1(k) + R_1 I(k)\left(1 - e^{\frac{-\Delta t}{R_1 c_1}}\right), \\ U_2(k+1) = e^{\frac{-\Delta t}{R_2 c_2}} U_2(k) + R_2 I(k)\left(1 - e^{\frac{-\Delta t}{R_2 c_2}}\right), \\ U_{\text{ocv}} = U(k) - I(k) R_{\Omega} - U_2(k) - U_1(k), \end{cases} \tag{1}$$

where Δt is the sampling period, $U_1(k)$ and $U_2(k)$ are the voltage values across the polarization resistance at time k, $I(k)$ is the current value at time k, $U(k)$ is the electromotive force of

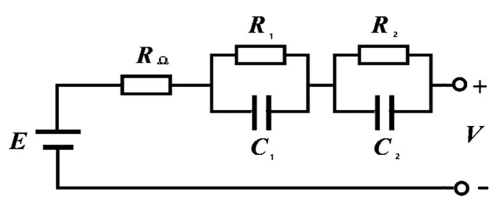

FIGURE 2 | Dual polarization equivalent circuit.

the battery at time k, U_{ocv} is the battery open-circuit voltage directly measured by the system, R_1 and C_1 are respectively the polarization internal resistance and capacitance characterizing the electrochemical polarization phenomenon, R_2 and C_2 are respectively the polarization resistance and capacitance characterizing the concentration polarization phenomenon, and R_Ω is the internal ohmic resistance.

As can be observed from Equation (1), several parameters in the battery state equation characterize the internal characteristics of the battery; for example, ohmic internal resistance R_1 and polarization internal resistance R_2 change with the battery charge/discharge rate, temperature, and charge/discharge cycle times [28]. These parameters change with the degree of battery degradation. Therefore, identifying these parameters and studying their change laws are necessary for developing SOC and SOH estimation algorithms.

To obtain various types of data, impedance curve fitting was performed in this study. First, the lithium battery was discharged at a rate of 0.5 C for 10 min, followed by a 10-min rest period; this process was repeated twice. The obtained voltage-time data were recorded and subsequently used for fitting the voltage curve for that period. According to the external characteristic model, the network response in the model represents a zero-state response. The voltage rebound curve (Figure 2) was fitted using Equation (2):

$$\begin{cases} \Delta U_1 = I_B R_\Omega, \\ \Delta U_2 = b_1 e^{-\tau_1 t} + b_2 e^{-\tau_2 t}, \end{cases} \tag{2}$$

where ΔU_1 represents the step change of the open-circuit voltage of the battery at the moment when the battery stops discharging. In practical applications, due to intermittent data sampling, it is expressed as the voltage rebound observed 0.5 s after the battery is disconnected. The parameters of feature matching, namely b_1, b_2, τ_1, and τ_2, were obtained by curve fitting using Equations (2)–(5):

$$b_i = R_i I, \tag{3}$$

$$\tau_i = \frac{1}{C_i R_i}, \tag{4}$$

$$R_\Omega = \frac{\Delta U_1}{I_B}. \tag{5}$$

2.3 | LSTM Neural Network With Attention Mechanism

Neural networks are well-suited for SOH estimation, particularly in complex, nonlinear environments. While conventional Backpropagation (BP) neural networks handle nonlinear inputs effectively, they struggle with temporal relationships in time-series data due to gradient issues [29–31]. In contrast, LSTM networks, designed for sequential data, overcome these challenges by capturing long-term dependencies and mitigating gradient problems. As depicted in Figure 3, this study enhances LSTM performance in SOH estimation by integrating a Multi-Head Attention mechanism, resulting in a robust LSTM-Attention model for SOC prediction.

2.3.1 | LSTM Cell Architecture

The LSTM cell architecture is built around three primary gates: the Forget Gate, Input Gate, and Output Gate, which

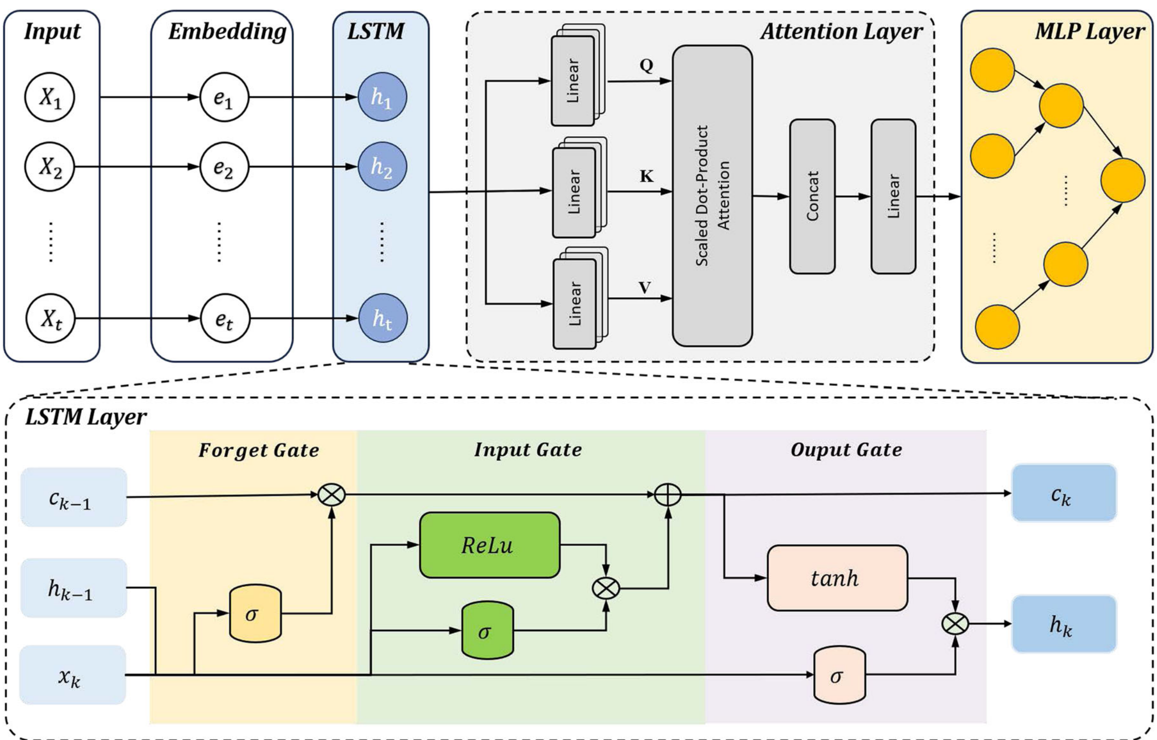

FIGURE 3 | Overall architecture of the improved LSTM neural network with multi-head attention.

collaboratively manage the flow of information to effectively capture long-term dependencies in sequential data. The Forget Gate determines how much past information should be retained, defined by the equation:

$$f_k = \sigma(W_f \times [h_{k-1}, x_k] + b_f), \tag{6}$$

where f_k modulates the previous cell state C_{k-1} based on the sigmoid-activated combination of the previous hidden state h_{k-1} and current input x_k. The Input Gate controls the integration of new information through the equations:

$$i_k = \sigma(W_i \times [h_{k-1}, x_k] + b_i), \tag{7}$$

$$g_k = \mathrm{ReLU}(W_C \times [h_{k-1}, x_k] + b_C), \tag{8}$$

where i_k and g_k determine the extent to which new data modifies the cell state, resulting in the updated cell state C_k as

$$C_k = f_k \cdot C_{k-1} + i_k \cdot g_k. \tag{9}$$

Finally, the Output Gate dictates the information passed to the next time step and the output, represented by

$$o_k = \sigma(W_o \times [h_{k-1}, x_k] + b_o), \tag{10}$$

$$h_k = o_k \cdot \tanh(C_k), \tag{11}$$

where h_k is the hidden state carrying the updated information forward. These gates work in unison, allowing the LSTM network to selectively retain, update, and output information, effectively addressing the challenge of learning long-term dependencies in sequential data.

2.3.2 | Multi-Head Attention

The integration of the Multi-Head Attention mechanism into the LSTM architecture enhances the model's ability to capture complex dependencies in sequential data by allowing it to focus on different aspects of the input sequence simultaneously. This combination leverages the strengths of both LSTM and attention mechanisms, enabling more effective time-series analysis, particularly in applications like SOH prediction.

The Multi-Head Attention mechanism operates by computing attention weights across various subspaces of the input data, as defined by the query Q, key K, and value V matrices. The attention output for a single head is formulated as

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \tag{12}$$

where d_k represents the dimension of the key vectors, and the softmax function ensures that the attention weights sum to 1. This mechanism allows the model to weigh the importance of different parts of the sequence effectively.

In the context of Multi-Head Attention, multiple such attention heads are utilized, each capturing distinct patterns within the sequence. The outputs of these heads are then concatenated and transformed linearly:

$$\mathrm{MultiHead}(Q, K, V) = \mathrm{Concat}(h_1, h_2, ..., \mathrm{h}_n)W_O, \tag{13}$$

where $\mathrm{h}_i = \mathrm{Attention}(QW_i^Q, KW_i^K, VW_i^V)$, and W_O is the linear transformation matrix applied to the concatenated outputs.

By incorporating this mechanism into the LSTM model, the combined architecture retains the sequential processing strengths of LSTMs while significantly enhancing its capacity to identify and prioritize critical information in the input sequence. This hybrid approach results in superior performance, particularly in tasks such as SOH prediction, where capturing complex temporal patterns is crucial.

2.4 | LSTM–AUKF Estimation Algorithm

The integration of LSTM networks with the AUKF marks a significant advancement in SOC estimation for battery management systems, addressing the inherent limitations of traditional Kalman Filters (KF). While KF is effective in state estimation by iteratively minimizing mean square error, it is limited to linear systems, making it less suitable for nonlinear dynamics. The AUKF overcomes this by employing the Unscented Transformation (UT), which better approximates the probability distribution of nonlinear functions, enhancing estimation accuracy, reducing computational complexity, and enabling real-time updates of noise statistics.

In this approach, the LSTM network first identifies key battery model parameters—ohmic internal resistance, polarized internal resistance, polarized capacitance, and maximum available capacity—through a discharge-static experiment. These parameters are then integrated into the AUKF framework, significantly improving SOC estimation accuracy and adaptability. Additionally, the multihead attention mechanism within the LSTM network allows the model to focus on the most relevant aspects of the input sequence, refining parameter estimation that feeds into the AUKF.

The combined LSTM and AUKF approach leverages the LSTM's predictive capabilities with the robust state estimation of the AUKF, effectively managing the inherent nonlinearities and uncertainties in battery dynamics. This dual approach leads to more accurate and reliable SOC predictions. The LSTM's ability to model complex temporal dependencies, combined with the AUKF's dynamic handling of noise and uncertainty, results in an advanced SOC estimation technique suitable for real-time battery management. This integration not only enhances SOC estimation precision but also ensures adaptability to varying operating conditions, optimizing battery performance and extending operational life.

The methodology involves several key steps, beginning with the initialization of the state vector and covariance matrix. The state vector x_0 and covariance matrix P_0 are initialized as follows:

$$x_0 = [\mathrm{SOC}_0, U_1, U_2]^{\mathrm{T}}, \quad P_0 = E\left[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)^{\mathrm{T}}\right], \quad (14)$$

where SOC_0 is the initial SOC, and U_1 and U_2 are state variables related to battery voltage. The initialized states are then used to predict the battery state:

$$f(x_k, u_k) = x^{k*} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \mathrm{e}^{\frac{-\Delta t}{R_1 C_1}} & 0 \\ 0 & 0 & \mathrm{e}^{\frac{-\Delta t}{R_2 C_2}} \end{bmatrix} x^{k-1} + \begin{bmatrix} \dfrac{\eta \Delta t}{C_{\mathrm{cap}}} \\ R_1\left(1 - \mathrm{e}^{\frac{-\Delta t}{R_1 C_1}}\right) \\ R_2\left(1 - \mathrm{e}^{\frac{-\Delta t}{R_2 C_2}}\right) \end{bmatrix} I_k + \omega^{k-1}, \quad (15)$$

where C_{cap} represents the current maximum battery capacity predicted by the LSTM network. Internal resistances R_1 and R_2, along with capacitances C_1 and C_2, accurately reflect the battery's dynamic behavior.

The AUKF then generates sigma points to propagate the state distribution through the system's nonlinear dynamics:

$$x_{k-1}^{i} = \hat{x}_0 \quad \text{or} \quad x_{k-1}^{i} = \hat{x}_{k-1} \pm \sqrt{(m + k_i) P_{k-1}}, \quad (16)$$

where m is the dimension of the state vector. The AUKF generates $2m + 1$ sigma points, which are crucial for capturing the distribution of the state vector as it evolves through the system's nonlinearities.

Next, appropriate weighting factors for the sigma points are calculated:

$$\begin{aligned} \omega_m^0 &= \frac{\lambda}{m + \lambda}, \\ \omega_c^0 &= \frac{\lambda}{m + \lambda} + (1 - \alpha^2 + \beta), \\ \omega_m^i &= \omega_c^i = \frac{1}{2(m + \lambda)}, \end{aligned} \quad (17)$$

where $\lambda = \alpha^2(m + \kappa) - m$, with $\alpha = 0.15$ and $\beta = 1.8$. These parameters help minimize higher-order errors in the state estimate.

The AUKF then updates the noise covariance matrices to account for system uncertainty and measurement noise:

$$\begin{aligned} Q(k+1) &= K(k+1) H(k+1) K(k+1)^{\mathrm{T}}, \\ R(k+1) &= \frac{H(k+1) + \sum_i \omega_c^i [Z^i(k+1) - y(k+1)]^2}{2}, \end{aligned} \quad (18)$$

where $H(k + 1)$ denotes the residuals and $K(k + 1)$ represents the gain matrix. This update is essential for maintaining accurate state estimates despite noise and uncertainties.

Finally, the AUKF iteratively refines the state estimates using the generated sigma points, calculated weights, and updated noise covariance matrices:

$$\begin{aligned} \hat{\mathrm{x}}(k+1|k) &= \sum_{i=0}^{2m} \omega_m^{(i)} \mathrm{x}^{(i)}(k+1|k), \\ P(k+1|k) &= \sum_{i=0}^{2m} \omega_c^{(i)} [\hat{\mathrm{x}}(k+1) - \mathrm{x}^{(i)}(k+1)]^2 \\ &\quad + Q(k+1). + Q(k+1). \end{aligned} \quad (19)$$

The AUKF then updates the estimated state values with new observation data:

$$\begin{aligned} K(k+1) &= \frac{P_{x_k z_k}}{P_{z_k z_k}}, \\ \hat{\mathrm{x}}(k+1|k+1) &= \hat{\mathrm{x}}(k+1|k) + K(k+1)[z(k+1) - \hat{z}(k+1)]. \\ &\quad \times [z(k+1) - \hat{z}(k+1)]. \end{aligned} \quad (20)$$

Through the overall iterative process shown in pseudocode, the LSTM–AUKF algorithm achieves high accuracy in SOC estimation, making it a powerful tool for real-time battery management applications. The integration of LSTM-derived battery parameters enhances the filter's adaptability to changing conditions, ensuring optimal battery performance and longevity.

ALGORITHM 1 | LSTM–AUKF Algorithm.

1: **Input:** Initial state x_0, covariance P_0, models $f(\cdot)$, $g(\cdot)$, LSTM parameters, measurements $z(k)$, sequence X

2: **Output:** Estimated state $\hat{x}(k)$, updated covariance $P(k)$

3: **Initialization:** Initialize x_0, P_0, Q_0, R_0, attention weights

4: **for** $k = 1$ to N **do**

5: Train LSTM on X to get hidden states h_t

6: Compute attention weights $\alpha_{t,i}$ for h_t, generate weighted context c_t

7: Apply multihead attention to c_t yielding c_t'

8: Predict state: $x_k^i = f(x_{k-1}^i, c_t') + \omega^{k-1}$

9: Compute predicted mean $\hat{x}(k|k-1)$ and covariance $P(k|k-1)$

10: Update using Kalman gain $K(k)$, update $\hat{x}(k|k)$ and $P(k|k)$

11: Update $Q(k + 1)$ and $R(k + 1)$ based on residuals

12: **end for**

13: **Final Prediction:** Use c_t' for final prediction

14: **Return:** $\hat{x}(k)$, $P(k)$

3 | Experiments and Results

In this section, we validate our algorithm through both Simulink simulations and practical testing platforms, presenting the details of the experimental setup and the results obtained. This process not only demonstrates the feasibility and effectiveness of the proposed algorithm but also provides a comprehensive insight into its performance in real-world scenarios.

3.1 | Test Platform

In industrial applications, lithium battery energy storage systems operate for extended periods amid high noise levels. Consequently, the BMS must possess robust anti-interference capabilities. Moreover, the EMF (Electrochemical Force)-SOC characteristic curve of a lithium battery clearly demonstrates that within the primary operational range of 5%–95% SOC, even minor measurement errors in the external battery voltage can lead to significant inaccuracies in SOC estimation. These inaccuracies may result in issues such as delayed equalization and erroneous system decisions. Therefore, to ensure the reliable acquisition of battery parameters necessary for validating our algorithm, the entire BMS must be capable of handling erroneous data while ensuring the collection of highly accurate data. In this study, we established a test platform to validate the functionality of the BMS and the accuracy of system parameter measurements. In addition, we conducted a series of tests to assess the performance of the system.

In this paper, Lithium Iron Phosphate (LiCoO2) batteries are selected as the cell type for the energy storage system, specifically the MAX-LOAD18650, which has a rated capacity of 2900 mAh, a nominal voltage of 3.7 V, and a discharge current of 0.58 A. These cells are designed to operate efficiently within a temperature range of −20°C to 60°C. After determining the type of battery, the design process involves tailoring to the overall parameter requirements of the energy storage system. Each sub-battery pack consists of 12 cells connected in series, and the number of sub-battery packs can be incrementally increased based on capacity needs. Detailed specifications of the battery packs are presented in Table 1 and the real-world test platform is shown in Figure 4.

The system adopts a one-to-many management approach, where each sub-battery pack is equipped with a battery management unit (BMU) module. This module connects the BMU's chip pins to the poles of each cell within the battery pack. To ensure the accuracy of data collection, this paper selects Linear's LTC6804-1 multi-cell battery stack monitor as the voltage monitoring chip, and designs the corresponding battery detection module, battery control module and the corresponding communication module. Additionally, a set of five negative temperature coefficient (NTC) thermistors are strategically placed across the battery pack to measure temperature at multiple points. The overall topology framework of the system is illustrated in Figure 5, showcasing a comprehensive strategy for monitoring and managing the battery packs to ensure optimal performance and safety.

TABLE 1 | Test sub-battery pack configuration parameters.

Items	Parameters
Cell specification	2900 mAh/3.7 V
Battery grouping method	1p12s
Battery pack capacity	34.8 Ah
Rated battery pack voltage	48 V
Voltage range	44.4–50.4 V
Maximum discharge current	0.6 A

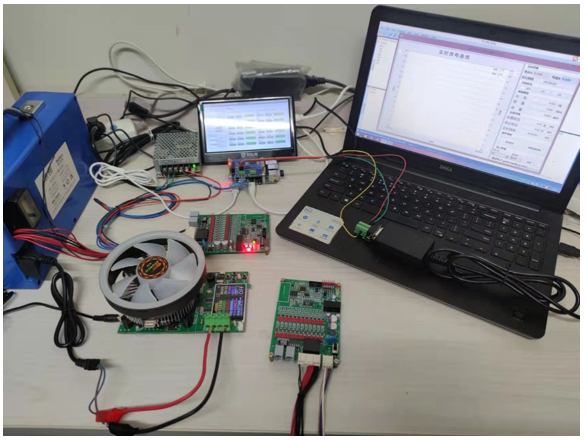

FIGURE 4 | Real-world test platform.

To verify that the experimental platform we developed meets the accuracy and real-time requirements for data acquisition, we conducted both static and dynamic voltage measurement accuracy experiments. In these experiments, we used constant current discharge at 3 and 0.3 A, and employed a high-precision multimeter to measure the voltage across each battery cell. The measured values served as reference points for comparison with the data obtained from the BMS. The voltage measurement module was employed to measure the voltage across each cell, and statistical analysis was performed on the resulting errors. The current control curve and the measurement error diagram are shown in Figure 6. Specifically, V_M0, V_M1, and V_M2 represent the measured voltages across the corresponding cells. As depicted in the figure, the voltage measurement error for a single cell remains within 1.5 mV, while the static voltage measurement error between 2 and 5 V is less than 2 mV, with an overall measurement error under 5%0. The system maintains high dynamic measurement accuracy under both high current (3 A) and low current (0.3 A) conditions, fulfilling the requirements for data set collection and algorithm validation.

3.2 | Simulink Model

In this study, an LSTM network model was constructed in MATLAB by using the aforementioned steps, including sample processing; the use of a training algorithm and an activation function; specifying the number of neurons in each layer; setting the weights, thresholds, number of training iterations, target values, and learning rate. Battery life is influenced by various factors, and although the input layer's node count does not affect the neural network's ability to discriminate between different inputs, it increases the duration of the learning process. To simplify and ensure the representativeness of input variables, we employed a three-layer forward network based on Kolmogorov's theorem to approximate continuous functions with arbitrary accuracy. Considering the charge and discharge characteristics of energy storage batteries, we used a three-layer neural network for SOH estimation. We set the input vector for the network's input layer as $[X_1X_2X_3X_4X_5]$. Where X_1 is the number of charging cycles, X_2 is the ambient temperature, X_3 is the ohmic internal resistance, X_4 is the polarization resistance, and X_5 is the polarization resistance. We set the number of output layer nodes (y) as 1, indicating the current maximum

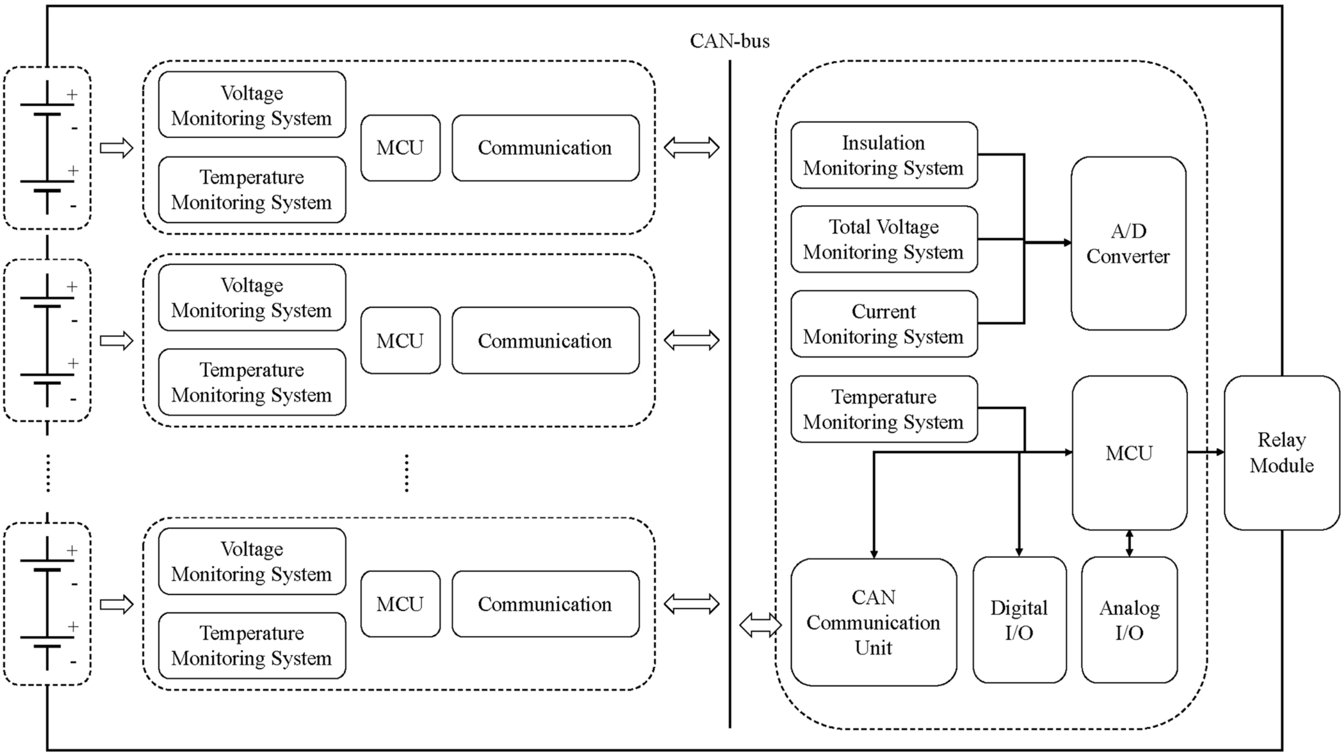

FIGURE 5 | Test battery platform topology.

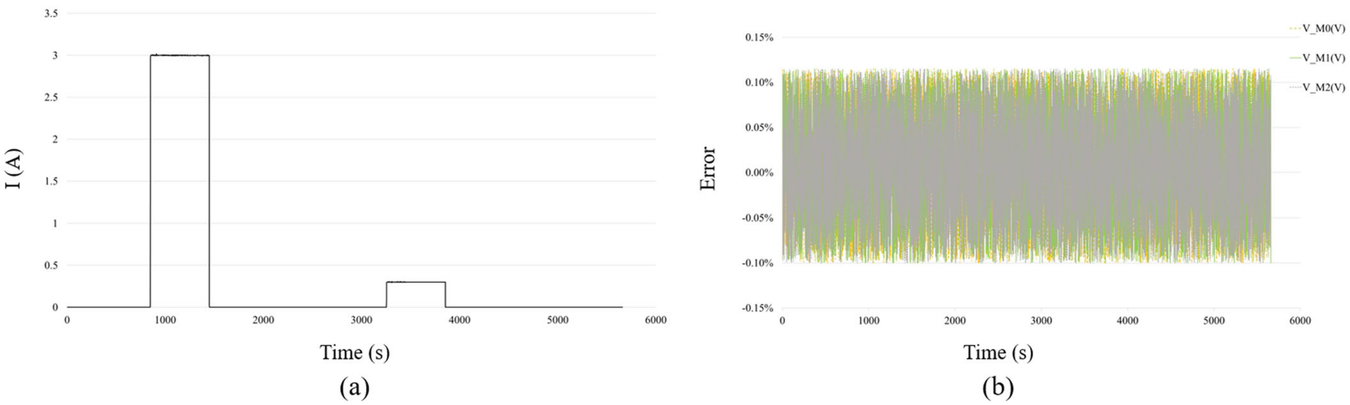

FIGURE 6 | Dynamic voltage measurement test.

available capacity C_{cap} of the battery. The cell structure of the LSTM layer is depicted in Figure 3. We adopted the Purelin linear activation function in the output layer. After several tests and adjustments, we found that the relationship between the input layer variables and the current maximum available capacity of the battery can be described accurately using 7 cells in the LSTM layer. The dimensions of the input data differ greatly. To avoid the effect of varying dimensions of input data, we standardized the output quantities R_Ω, R_1, R_2, T, N, and C_{cap} by using the following formula:

$$X_s = \frac{X - X_{\min}}{X_{\max} - X_{\min}}, \tag{21}$$

where X_s is the normalized value, $X_{\max}$ is the maximum value in the data sequence, and $X_{\min}$ is the minimum value in the data sequence. We constructed the test simulation model (Figure 7) in Simulink.

In Figure 7, module N_LUT is the input variable; module Em_LUT, SOC_LUT, Cap_LUT, and $Temperature\_LUT$ are the data sets obtained from the SOC-OCV characterization experiment; and module R_0_LUT (representing ohmic resistance), module R_1_LUT, and module R_2_LUT are the data sets obtained by parameter identification tests.

3.3 | Network Training Details

To evaluate the performance of the proposed model on a newly tested battery, galvanostatic charge-discharge experiments and battery impedance characteristic tests were conducted under

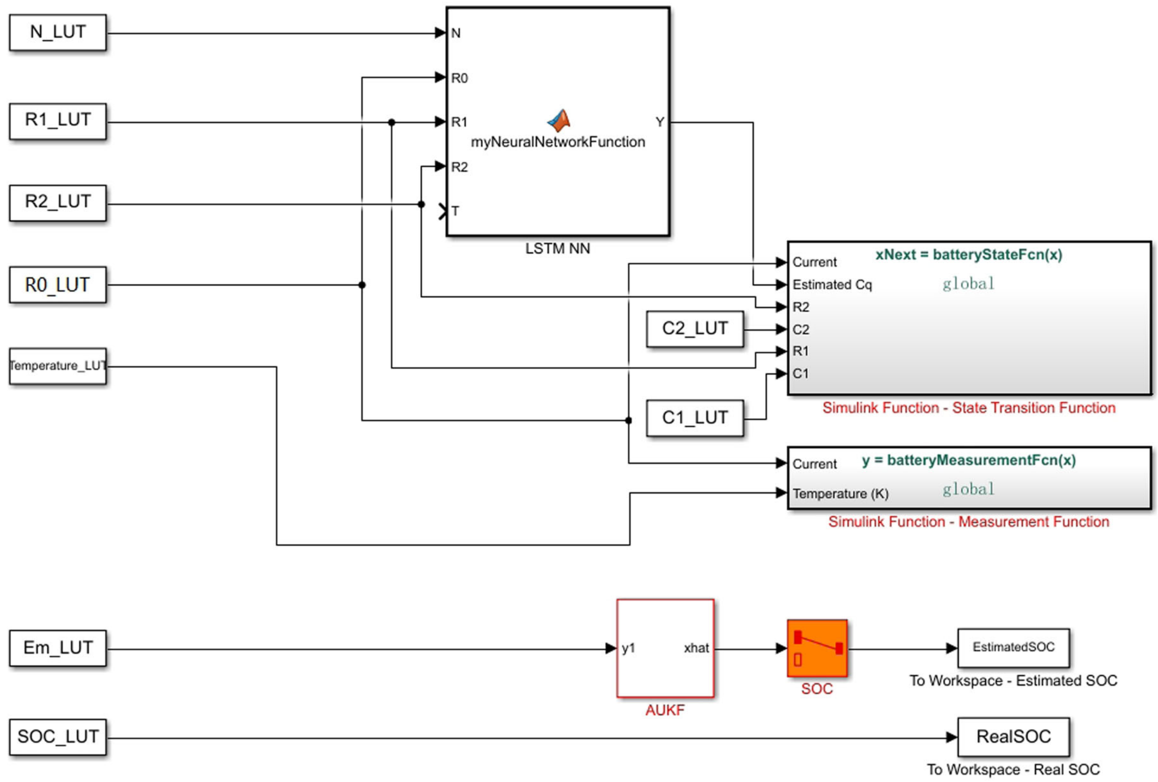

FIGURE 7 | Simulink model.

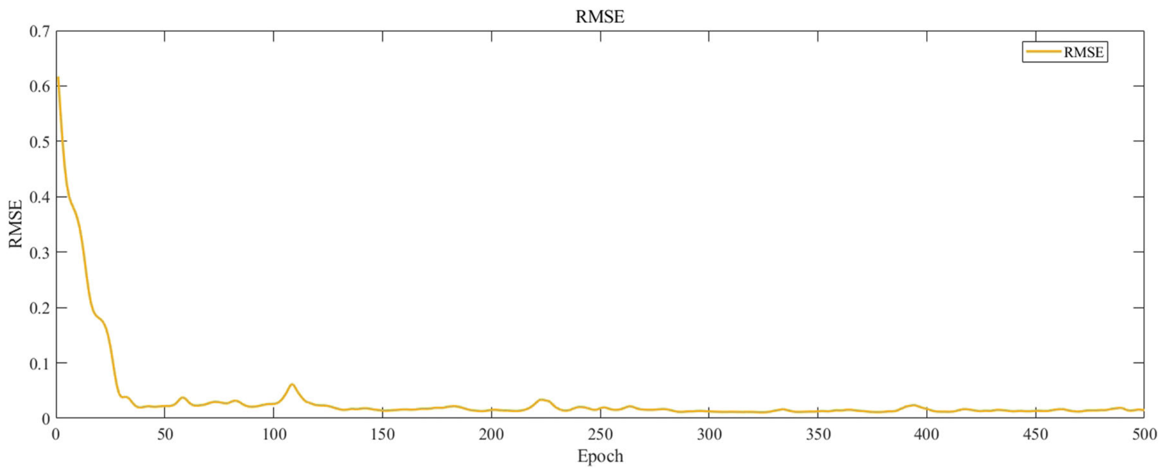

FIGURE 8 | RMSE of training process.

standard room conditions until the battery reached a predefined failure criterion, defined as a reduction in capacity to 70% of its rated value. During the charging and discharging cycles, real-time power and voltage data were systematically recorded for each cell. This process resulted in the creation of an aging data set comprising 12 battery cells, capturing their states from initial operation to failure. The data set was then divided into two subsets: 8 samples for model training and 4 samples for performance testing. The network was trained using the "train network" function with a maximum of 500 training epochs and default parameters. We used the root mean square error (RMSE) and loss as objective functions; their convergence is illustrated in Figures 8 and 9, respectively. The RMSE of the test sample prediction results was 0.052, and the loss was less than 0.002.

To further validate the robustness of our model, we designed an experiment where the test data was intentionally modified to introduce a ten percent error margin as input. This methodology aimed to observe the model's performance in managing initial inaccuracies. By integrating this error into our evaluation process, we sought to simulate real-world conditions where data might not be perfect, thus providing a more comprehensive understanding of the model's adaptability and resilience in handling deviations from expected inputs. The trained LSTM-Attention neural network model was imported into Simulink,

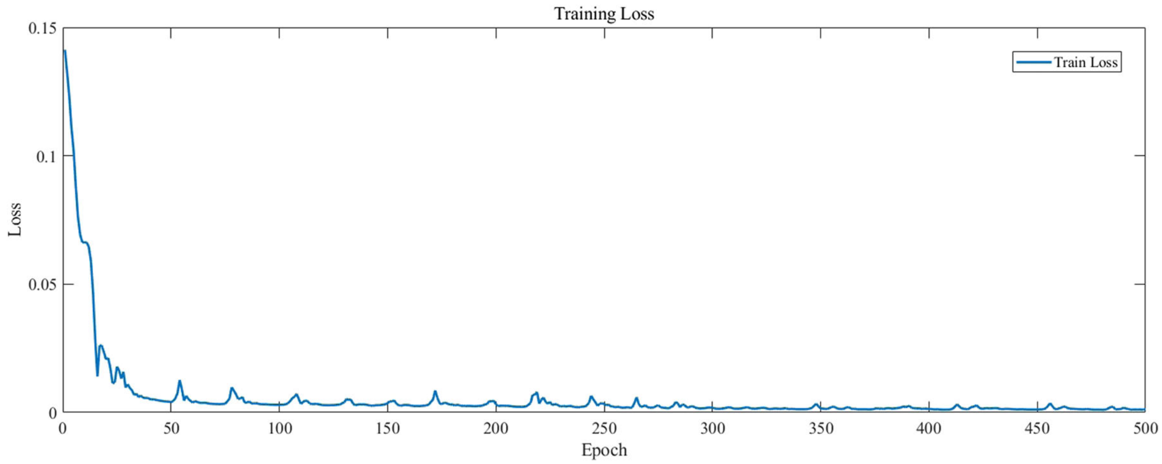

FIGURE 9 | Training loss.

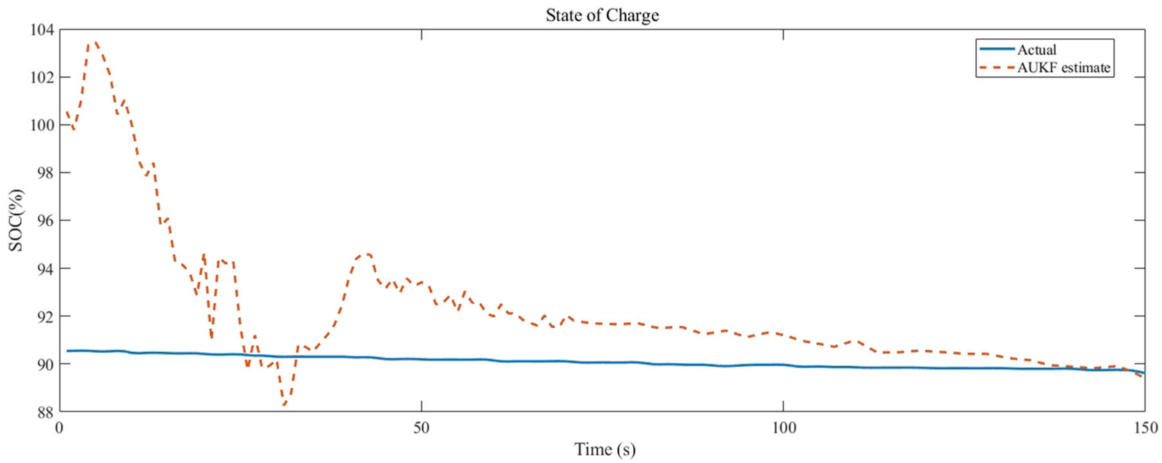

FIGURE 10 | Comparison of LSTM–AUKF model test samples and predicted values with 10% initial set error.

and the estimated SOC values were compared with the experimentally measured SOC values. To evaluate the accuracy and robustness of joint SOC-SOH estimation model, we set the initial SOC error as 10% for the test samples. A comparison of the estimation results with actual values is shown in Figure 10, whereas the model's estimation error is shown in Figure 11. The error converged quickly, with the estimation error within 1% after 100 cycles, thus indicating an effective correction of the initial error.

To demonstrate the superiority of our model, we simultaneously trained a back propagation-adaptive unscented Kalman filter (BP-AUKF) combined algorithm and an attention-based model (CNN-BiLSTM-Attention, i.e., Convolutional Neural Networks, Bidirectional Long Short-Term Memory, and Attention Mechanism) using the aforementioned 8 sets of cell aging data. For comparison, we evaluated the trained models on the remaining 4 data sets, and the experimental results are presented in Figure 12. Performed on the Simulink simulation platform, the prediction error for SOH estimation using the BP-AUKF algorithm was 0.53%, while the CNN-BiLSTM-Attention model achieved a prediction error of 0.41%. In contrast, the LSTM–AUKF algorithm demonstrated a significantly lower prediction error of 0.34%. This comparison highlights the superior accuracy and efficiency of the LSTM–AUKF algorithm in predicting SOH, thereby confirming its enhanced performance over both the BP-AUKF and CNN-BiLSTM-Attention algorithms in the context of this study.

3.4 | Real-World Test

Although the proposed model exhibited good performance in network training, to test its practical applicability, we deployed the model on real-world devices by using a fully charged battery pack. Subsequently, we discharged the battery pack by using constant current discharge method until it was completely depleted, resulting in a depth of discharge of 100%. We closely monitored the data to detect any error accumulation. As indicated in Table 2, we compared the final experimental outcomes with existing predictive methods, including model-based, data-driven, and hybrid approaches. This comparison specifically emphasized the contrast in SOH prediction results, noting that recent studies have maintained SOC prediction errors within the range of 1.5%–2%, a performance bracket within which our model also falls. The findings indicate that our model achieved a higher level of predictive accuracy, thereby affirming the effectiveness of the proposed framework. This accomplishment

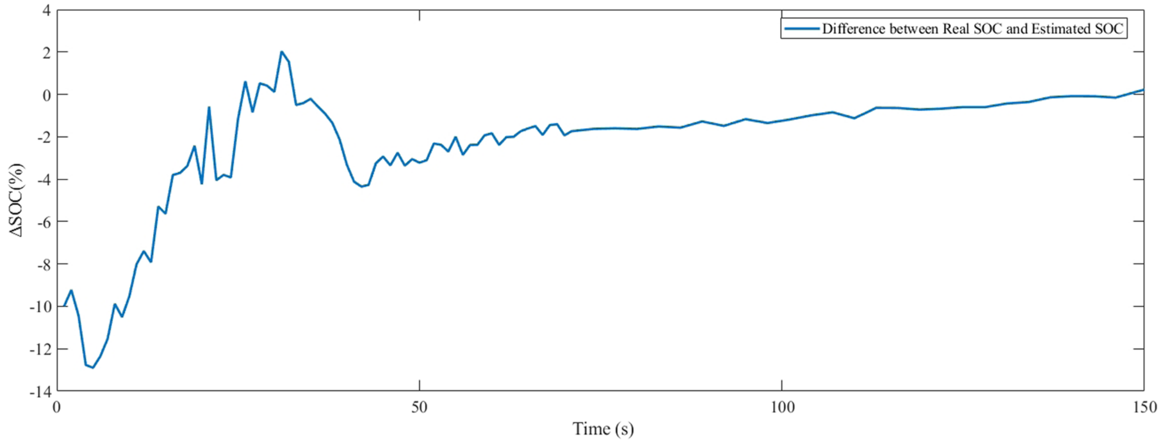

FIGURE 11 | Error diagram of the LSTM–AUKF model in Simulink with 10% initial set error.

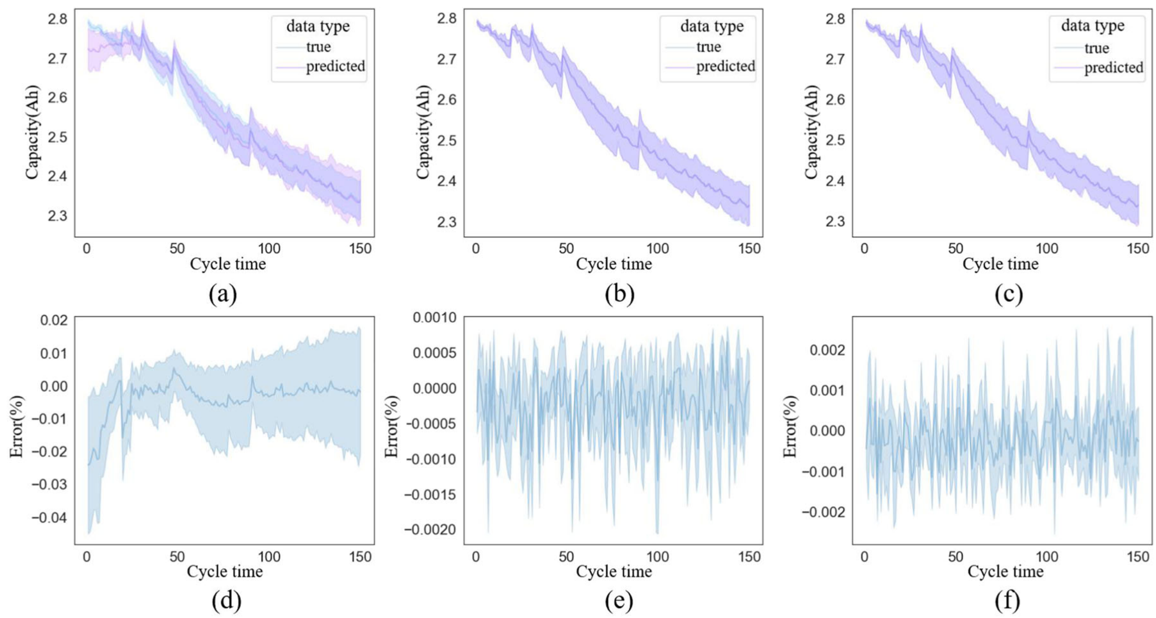

FIGURE 12 | Comparison between BP-AUKF, CNN-BiLSTM-Attention and LSTM–AUKF models, (a) and (d) denote the prediction results as well as the prediction error of the BP-AUKF algorithm for SOH; (b) and (e) denote the prediction results as well as the prediction error of the CNN-BiLSTM-Attention for SOH; (c) and (f) denote the prediction results as well as the prediction error of the LSTM–AUKF with Attention Module for SOH.

evidence our model's capacity to successfully address the challenges outlined in the first section, underscoring its potential to make a significant contribution to the advancement of predictive analytics. Through this comparative analysis, our research not only showcases the superior accuracy of our model but also validates the innovative approach of blending various predictive methodologies to enhance prediction precision.

Utilizing the test battery packs outlined in the preceding sections, our research team executed practical evaluations of the algorithm under controlled experimental conditions at temperatures of 25°C, 15°C, and 5°C. Each experimental condition was subjected to three rounds of testing, culminating in a total of nine distinct tests. To facilitate a more intuitive interpretation of the data, we averaged the SOC outcomes of the battery packs and juxtaposed these values with the estimated errors derived from the algorithm. This comparative analysis, aimed at elucidating the precision of the algorithm in varying thermal contexts, is visually detailed in Figure 13.

The experimental results presented in Table 3 elucidate the impact of temperature on the predictive accuracy of the fusion algorithm discussed in this study. At a temperature of 25°C, the algorithm achieved its highest predictive precision, with an accuracy of 1.277%. Conversely, at 5°C, the algorithm's accuracy diminished, resulting in a higher error rate of 2.095%. Across all three temperature conditions tested—25°C, 15°C, and 5°C—the algorithm maintained an average predictive accuracy of 1.715%. This demonstrates that despite the variations in temperature, the fusion algorithm exhibits a commendable level of predictive accuracy under diverse conditions. The findings highlight the algorithm's robustness and its capability to deliver reliable predictions in varying thermal environments. This adaptability underscores the effectiveness of the fusion approach in predictive models, suggesting its

TABLE 2 | Comparison of the latest SOC and SOH estimation methods and proposed framework.

Methods	Model	SOH prediction error
Open circuit voltage [32]	Model-based	5%
Dual extended Kalman filter [33]	Model-based	3%
Unscented particle filter [34]	Model-based	2.5%
Novel health indicator [35]	Data-driven	3%
Ensemble learning [36]	Data-driven	2%
Support vector regression [37]	Data-driven	1.5%–2%
Reinforcement learning [38]	Data-driven	1.5%
Extreme learning machine [39]	Data-driven	2%
IC-BP [40]	Number-model fusion	1.16%
NSSR-LSTM [41]	Number-model fusion	0.8%
LSTM [42]	Data-driven	0.6%
BP-AUKF	Number-model fusion	0.68%
CNN-ASTLSTM [43]	Data-driven	0.54%
Attention-based CNN-BiLSTM [44]	Data-driven	0.51%
Proposed LSTM–AUKF without attention module	Number-model fusion	0.5%
CNN-BiLSTM-Attention [45]	Data-driven	0.54%
CNN-LSTM-Skip [46]	Data-driven	0.46%
AM-seq2seq [47]	Data-driven	0.43%
Proposed LSTM–AUKF	Number-model fusion	0.39%

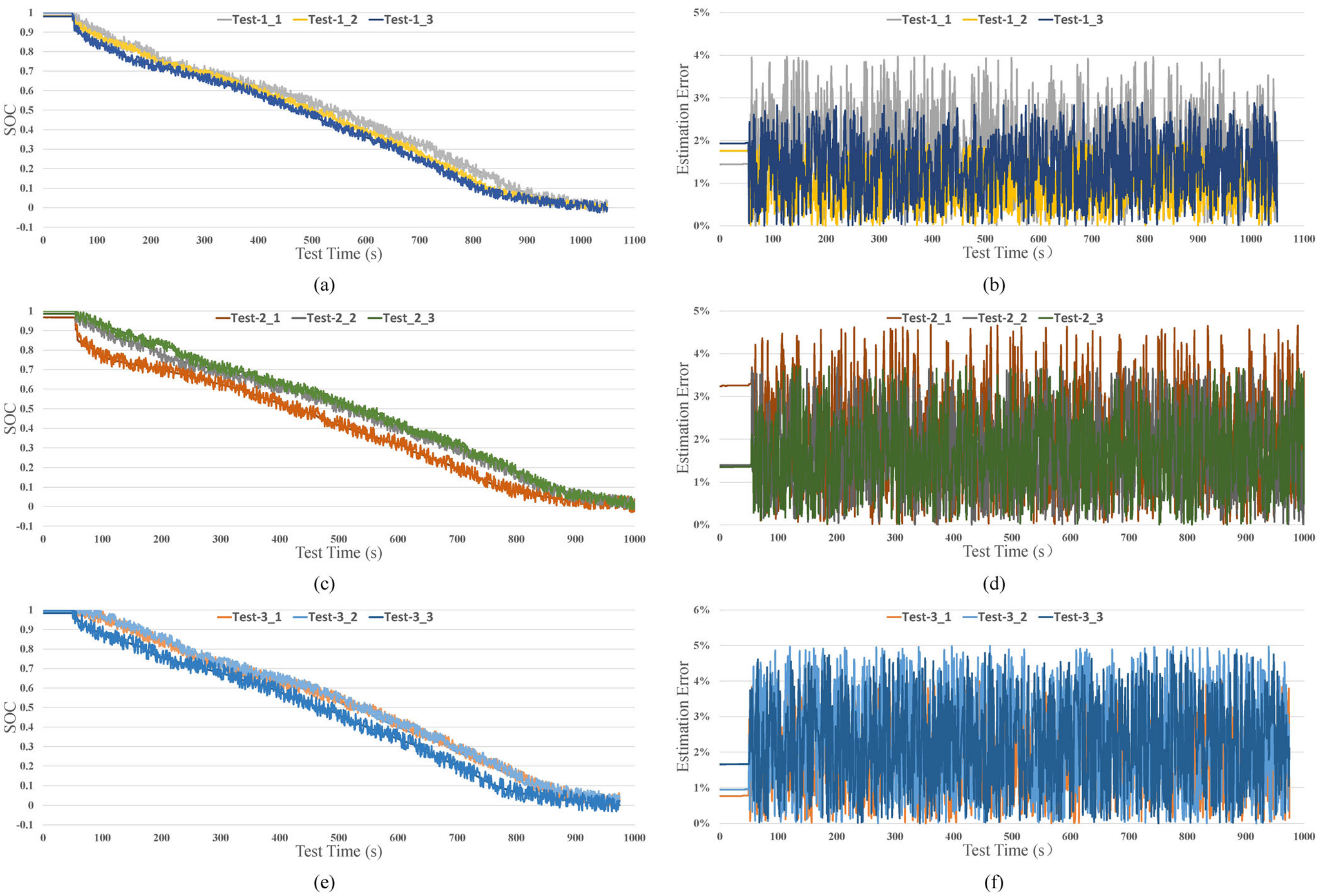

FIGURE 13 | Comparison of estimated and actual SOC values and error diagrams. (a) and (b) show the tests under 25°C, (c) and (d) show the tests under 15°C, (e) and (f) show the tests under 5°C.

TABLE 3 | Numerical results of SOC prediction.

Test temperature		25°C			15°C			5°C	
Estimation MAE	1.387%	1.357%	1.089%	1.641%	1.845%	1.832%	2.320%	2.003%	1.962%
Average		1.277%			1.772%			2.095%	

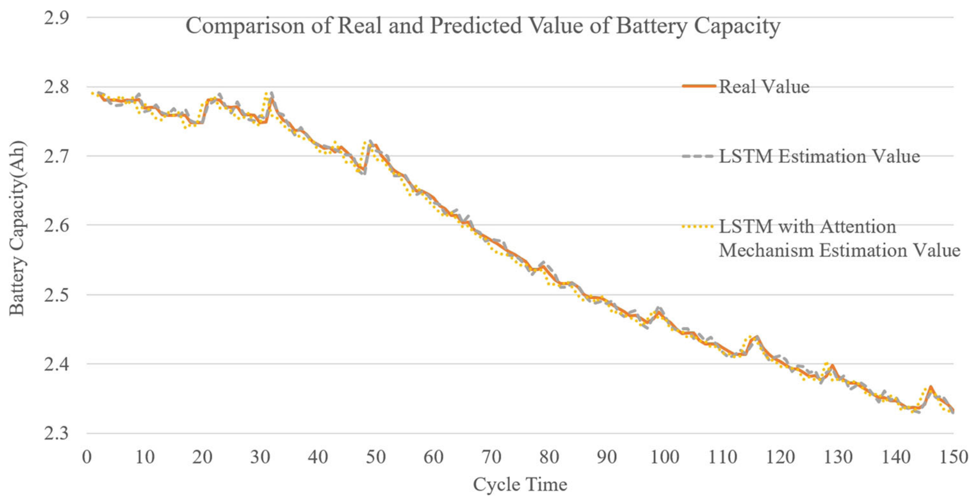

FIGURE 14 | Comparison of real SOH and predicted SOH values of each cell.

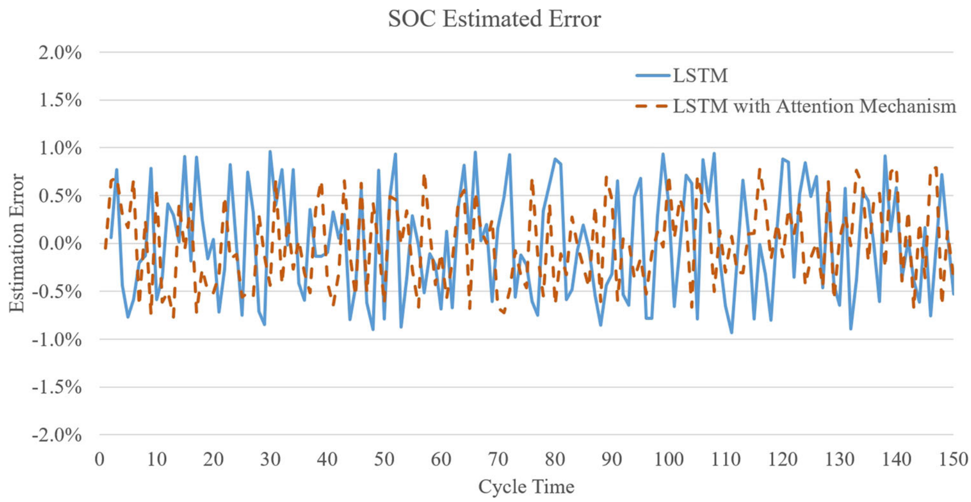

FIGURE 15 | Estimation error of SOH.

potential applicability in scenarios where environmental conditions may vary significantly.

Furthermore, we assessed the accuracy of the algorithm by continuously charging and discharging the battery packs until their capacity reached 70% of the maximum capacity, which signifies battery failure. We conducted twelve sets of tests; the average of the results is shown in Figures 14 and 15. In the study conducted on battery characterization measurements utilizing an experimental platform, the implementation of a novel LSTM–AUKF prediction algorithm was scrutinized for its effectiveness in forecasting the SOC and SOH of batteries. The experimental results clearly indicate that our algorithm exhibits a high degree of accuracy throughout the entire evaluation process. Remarkably, even without the attention module, the model achieves prediction errors of just 1.715% for SOC and 0.503% for SOH. The inclusion of the attention module further refines the model's performance, reducing the SOH prediction error to 0.39%, as demonstrated in Table 2, underscoring its superiority over other algorithms. This innovative approach employs the integration of AUKF parameters with neural network forecasts, significantly enhancing prediction reliability to maintain errors under 2% for both SOC and SOH indicators. Notably, the algorithm demonstrated a comparatively moderate overall prediction error, showcasing its enhanced capability in accurately predicting SOH over SOC. This underscores the potential of combining LSTM neural networks with AUKF for improving predictive accuracy in battery health and charge level assessments.

4 | Conclusions

The exploration and development of novel methods for estimating the SOC and SOH of batteries are crucial in advancing BMS and enhancing the efficiency and longevity of energy storage solutions. This paper presents a groundbreaking approach by integrating a dual-polarized equivalent circuit model with advanced machine learning techniques, specifically an LSTM neural network and an AUKF, to achieve this goal. The hybrid methodology proposed herein not only addresses the inherent challenges associated with SOC and SOH estimation but also introduces a sophisticated mechanism capable of coping with the noise and nonlinearity that typically impair accuracy in real-world settings.

The incorporation of the fusion model consisting of AUKF and LSTM network is particularly noteworthy, as it dynamically adjusts the statistical properties of noise and learns features from the data, thereby significantly mitigating its impact on the estimation process. This feature is instrumental in enhancing the reliability and precision of SOC and SOH estimations, even in environments characterized by high levels of noise and unpredictability. Through rigorous evaluation on an experimental platform, the algorithm's performance has been validated, showcasing its exceptional capability in accurately predicting battery health. Such achievements underscore the practicality and innovative perspective brought about by the proposed method in the realm of battery SOH and SOC estimation.

One of the salient benefits of the proposed approach lies in its flexibility and scalability, which allow it to adeptly process real-time data from various energy storage systems. This adaptability facilitates the automatic extraction of operational characteristics of batteries, thereby significantly improving the accuracy of health estimation and, by extension, the robustness of the overall BMS. The development of this joint estimation algorithm represents a significant stride toward optimizing battery performance and longevity, contributing to the sustainability and efficiency of energy storage technologies.

However, the study also candidly acknowledges certain limitations that underscore the need for further exploration and refinement. A significant area yet to be explored is the algorithm's performance in scenarios involving large-scale battery packs, a common configuration in many practical applications ranging from electric vehicles to grid storage solutions. This gap in the research suggests that while the algorithm shows potential, its scalability and applicability in larger, more complex systems remain uncertain. Moreover, the algorithm's ability to maintain accuracy in SOC and SOH predictions during system failures presents another critical avenue for investigation. Ensuring resilience and reliability in adverse conditions is paramount for battery management systems, particularly in applications where safety and operational integrity are of utmost importance. These identified limitations not only highlight the necessity for comprehensive testing and research but also open up opportunities for enhancing the algorithm's design and expanding its applicability in real-world scenarios, thereby contributing to the advancement of battery technology and energy management systems.

Acknowledgments

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